

I develop UniversalToolchain/Wist2: typed cross-package language plans, exact manifest binding, callable-first SSA, the public Wist facade, and PlanFuzz for configuration-aware differential testing.

Typed language plans

routes, passes, capabilities, exact executors

Callable-first SSA

descriptor semantics and verifier-gated roundtrip

1,459 / 1,459

0 failures; nine packages and clean consumers

EXPERIENCE

2024 — PRESENT	UniversalToolchain / Wist2 - Typed SDK for independent language packages, deterministic routes/pass ordering, and schema-v5 lock identity. - Fail-closed runtime policies, exact manifest binding, and lifecycle ownership. - Callable-first SSA and PlanFuzz: seven oracle families, replay, exact fingerprints, and reducers.
JUL–AUG 2026	MCST · LLVM track - C++23 graph-analysis core: RPO/cycles, Dijkstra, Tarjan SCC, and Dinic. - CMake, warnings-as-errors, ASan/UBSan, clang-tidy, and I/O tests.

SELECTED PROJECTS

PlanFuzz Configuration-aware differential testing with fresh-process confirmation and deterministic program/plan reduction.
PS-form Analyzer Ranked 1st of 49 with 104/104 tests; conservative memory-dependence analysis and an exact-oracle harness.
x86-64 Codegen Lab SSA-like IR, liveness, linear scan, iced-x86 emission, and differential validation.

SKILLS

Language architecture typed artifacts, contribution graphs, routes, passes, manifests
IR / SSA Bytecode/AIR, callable descriptors, CFG, dominance, folding, SCCP-lite, DCE
Verification differential/metamorphic testing, exact oracles, replay, reducers, parity
Languages C#/.NET, C++23, C17, Python, Rust

EDUCATION

Software Engineering student at HSE University

RECOGNITION

Ranked 1st of 49 in MCST selection; HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026; Baltic Grand Prize.

TECHNICAL COMMUNICATION

About 50 students; practical NASM IA-32 course; public documentation and technical writing.

AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026